

Crystallization of GST225 thin film induced by a single femtosecond laser pulse: Experimental and theoretical study

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ABSTRACT

Amorphous to crystalline phase transition in $\text{Ge}_2\text{Sb}_2\text{Te}_5$ film under the influence of a single femtosecond laser pulse is studied. Two-dimensional temperature calculations and kinetic model for crystallization were used to support experimental results and then to explain the fast mechanism of crystallization. Based on comparison with the experimental data, the theoretical Time-Temperature-Transformation diagram was calculated, that allowed to define the range of cooling rates at which crystallization is possible. The distribution of crystalline fraction in the thin film was calculated using these rates. Reflectance of the simulated structures turned out to be in good agreement with experimental observations.

1. Introduction

The telluride based alloys, such as $\text{Ge}_2\text{Sb}_2\text{Te}_5$ (GST225), possess a unique combination of structural, electrical, and optical properties. The amorphous and crystalline states of these materials exhibit a significant difference in optical reflectivity and electrical resistivity, while the transition between these two phases may be achieved reversibly on very short timescales [1].

This set of properties attracts a lot of interest to the GST225 within the field of optical data storage [2] and non-volatile memories [3,4]. The basic idea of such data storage applications relies on the reversible transformation between the disordered amorphous and ordered crystalline states, which in particular can be achieved by local influence of short laser pulses. In terms of efficiency of such devices, the important issues appear to be the nature of laser-induced phase transitions in GST225 thin films and methods of increasing the transformation rate. One of the possible solutions is to use ultra-short laser pulses. Phase transitions in GST225 thin films caused by an ultrafast (femtosecond or picosecond) pulsed laser are of great interest in the field of phase change memory (PCM) materials.

Crystallization triggered by ultra-short laser pulse has already been

reported for the Ge-Sb-Te materials of various compositions. Thus, crystallization of $\text{Ge}_1\text{Sb}_2\text{Te}_4$ 100 nm thick film has been achieved by a proper control of the film thickness with a wavelength 800 nm [5]. The $\text{Ge}_2\text{Sb}_2\text{Te}_5$ film of 90 nm thickness has been crystallized by a single ultraviolet laser pulse (wavelength $\lambda = 248$ nm, pulse duration $\tau = 500$ fs) [6]. In addition, crystalline structure was observed after irradiation of $\text{Ge}_2\text{Sb}_2\text{Te}_5$ films with $\tau = 120$ fs pulse of $\lambda = 800$ nm [7]. Conversely, crystallization of amorphous material was not achieved by a single laser pulse with the same parameters as reported by authors [8].

In our previous study [9] we did not achieve single pulse crystallization as well. So, in spite of the efforts made in the abovementioned works, the question remains to be open: what conditions must be created for crystallization by a single ultra-short laser pulse becomes possible?

The crystallization induced by a laser pulse is regarded as a non-isothermal process with non-Arrhenius behavior and is characterized by high rates of material temperature changes [10]. The effect of heating rate on the crystallization temperature has been studied by Choi et al. [11]. As a result, the empirical formula has been proposed, according to which the crystallization temperature becomes unexpectedly high at great heating/cooling rates (~ 1 K/ns). The results have been confirmed by Burr et al. [12]: according to their simulation at an extremely great

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heating rate the crystallization temperature shifts to 550–600 K from the value 423 K which is determined from the conventional DSC (at 10 K/min) [13].

In the given paper, we study amorphous to crystalline phase transitions in $\text{Ge}_2\text{Sb}_2\text{Te}_5$ under the influence of a single femtosecond laser pulse. Specifically, we aim to define the conditions that should be created to achieve single-pulse crystallization of this material. Based on experimental data, we developed a numerical model that distinguishes the role of the ultra-fast heating/cooling rate for this phase transition.

2. Material and methods

2.1. GST225 thin film

Amorphous GST225 films of 230 nm thicknesses were deposited by direct current magnetron sputtering of a crystalline target (ACI alloys) on the substrates with contrasting thermal conductivity and optical properties. Two types of substrates were used: thermally oxidized silicon wafers (thickness of wet thermal oxide is 1 μm) and the same wafers, coated consistently with TiN sublayer (50 nm) and tungsten layer (250 nm). Further in the text, for clearness, given substrates will be marked as “substrate I” and “substrate II”, correspondingly. The pressure of Ar^+ ions during the process of GST225 film deposition was 0.57 Pa; the sputtering power was 100 W. The surface roughness, phase state and distribution of elemental composition for the as-deposited films were controlled by atomic-force microscopy, X-ray diffraction analysis and Auger electron spectrometry respectively as described in in Ref. [14].

2.2. Laser irradiation

The irradiation was performed with single femtosecond pulses from Yb:KGW laser system Pharos SP (Light Conversion) at a wavelength of 1030 nm. The pulse duration was determined with a pulse Check autocorrelator (APE) and was equal to 185 fs. The radiation was linearly polarized, focusing was carried out by a lens with a focal length of 75 mm (Thorlabs, LA1608-B); the spot diameter on the surface of the GST layer was about 60 μm ($1/e^2$).

The sample was placed on the ABL1000 air-bearing XYZ-motorized stage (Aerotech) synchronized with the femtosecond laser. The stage translation and laser shots were programmed using SCA Professor software (Workshop of Photonics). The pulse energy was controlled with the WattPilot attenuator (Altechna).

The sample was shifted along the line in the plane orthogonal to the optical axis. Every 200 μm , the stage was stopped and a femtosecond pulse was delivered to it, then the sample was moved to the next point, at which it was irradiated again but with a pulse of different energy. Thus, the sample was irradiated with single pulses of energy in the range 200 ÷ 1200 nJ with a step of 50 nJ. Three lines of laser spots were recorded with identical irradiation regimes and identical results were obtained for the pulses of the same energy demonstrating good reproducibility of the experimental setup.

3. Results

Earlier, using the same laser system, we have showed that the phase transformation of films less than 130 nm thickness can be triggered by applying multiple femtosecond laser pulses only [9]. The required number of pulses causing the crystallization decreased with the film thickness increasing (Fig. 1). Moreover, this number of pulses did not depend on the substrate conductivity type (the same substrates were described in Section 2.1; their conductivity values are presented in Table 1). This regularity suggests that, the films of a sufficiently large thickness could be crystallized with a single femtosecond pulse. Thus, the aim of the current study becomes to approve this hypothesis.

Irradiation with an insufficient number of pulses with energies up to 500 nJ formed so-called “pale spots” with optical contrast (OC) about

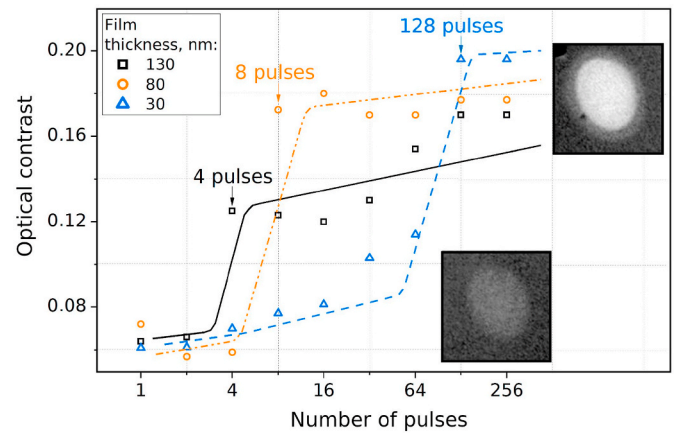


Fig. 1. The abrupt growth of optical contrast with an increase of the number of pulses observed for the films with different thickness; pulses energies were 500, 350 and 250 nJ for 130, 80 and 30-nm films. Optical images of the modified areas obtained on 130 nm thick film on substrate I after irradiation with 1 (bottom image) and 4 (top image) pulses are presented as an example of the pale and bright spots, correspondingly.

Table 1

Parameters of amorphous GST225 and substrates used in modeling.

k – thermal conductivity, mW/($\mu\text{m K}$)	0.25×10^{-3} [23,24]
c – specific heat, mJ/(g K)	212 [25]
ρ – density, g/ μm^3	5.87×10^{-12} [26]
R – reflection coefficient (for wavelength of 600 nm)	0.428 [22]
α_{eff} – absorption coefficient (for 1030 nm wavelength), μm^{-1}	16.2
T_m – melting temperature, K	880 [27]
T_g – glass transition temperature, K	420 [28]
η_{∞} – extrapolated infinite temperature viscosity, Pa $\times$ s	1.1×10^{-5} [29]
m – fragility index	67
ΔH_f – enthalpy of fusion, J/cm 3	670 [30,31]
σ – interfacial energy, J/m 2	0.063 [12,32]
E_{η} – viscosity activation energy, eV	2 [33,34]
$k_{\text{substrate}}$ – substrate thermal conductivity (W/SiO $_2$), mW/($\mu\text{m K}$)	163×10^{-3} [35] 1.45×10^{-3} [35]
α_s – substrate thermal diffusivity (W/SiO $_2$), $\mu\text{m}^2/\text{s}$	6.3×10^7 [35] 9.2×10^5 [35]

5% only (see the lower inset in Fig. 1). A further increase of the pulse energy leads to ablation of the material. Appearance of the pale spots is an indicator of the ongoing restructuring, and earlier we assumed that a transition from an amorphous to a glassy state as the result of melt-quench process was observed.

It was demonstrated that in the case of 230 nm thick film a single femtosecond laser pulse triggered the great change of reflectivity. Optical images of modified areas, obtained on the film on a dielectric substrate, are shown in Fig. 2 (top row). When irradiated with a pulse of energy less than 350 nJ, no visible changes are observed inside the irradiated area. Influence of 350 nJ pulse leads to the appearance of a pale spot. At 450 nJ the formation of a brighter center can be detected. Further increase of the pulse energy is accompanied with growth of the radii of the both pale and bright areas. As a result, the bright spots surrounded by a pale ring are formed (Figs. 2, 650 nJ). Optical contrast of pale rings with amorphous background is less than 5% and does not depend on the pulse energy. Optical contrast of bright spots, on the contrary, grows with the pulse energy increasing (see bottom row in Fig. 2). Qualitatively, the same results were observed in the case of the conductive substrate. The only difference is the threshold energies: the

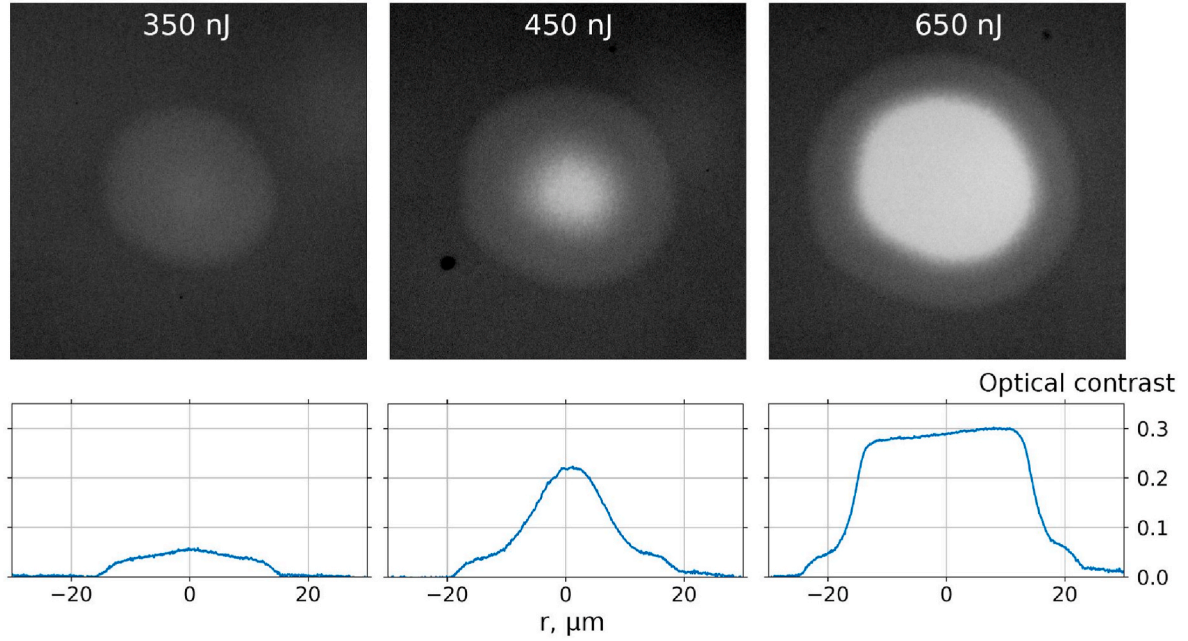


Fig. 2. Optical images of the modified areas (top row) and corresponding optical contrast profiles (bottom row), for the film on the substrate I.

pale spots appeared under irradiation with pulse 400 nJ and more, and the bright center started to form under irradiation with 500 nJ pulse.

We assumed that the temperature regime required for crystallization is achieved only at a sufficiently large film thickness. The driving force for crystallization is proportional to undercooling. Therefore, in this case the crystallization is initiated only when the material, initially melted by laser pulse, is cooled below its melting point due to heat dissipation. The description of the crystallization process in fact is equivalent to simulation of the quenching process. While molten material is cooled down, a competition between two processes occurs: a decrease in the atomic mobility and formation of the crystalline phase, specifically, discrete nuclei. Nucleation is not instantaneous, so that for the crystallization proceeding it is necessary to keep the material for sufficient time under elevated temperature, when increased atomic mobility facilitates atomic rearrangement. If time for material solidifying is shorter, virtually no crystalline phase is formed, and the glassy phase formation is observed. That is why it is commonly accepted that the formation of glass upon the melt-quenching process is governed by the cooling rate. This idea appears to be the main point for considering the quenching process using Time-Temperature-Transformation (TTT) diagrams [15,16]. A TTT diagram is constructed by calculating the time that is necessary to transform a particular fraction of material to the crystalline phase at a given temperature.

Thus, to describe the crystallization process triggered with a single femtosecond pulse, we estimate the heat distribution in a film caused by the laser irradiation, its subsequent dissipation, and phase transformations of the film material accompanying these processes.

4. Simulation of laser-induced phase transitions

In order to develop a mathematical model of the observed laser-induced modification of GST225, three processes should be accounted for attention. The first one is the specific influence of laser pulse on the crystallization. We suppose that in our experiment the leading role is played by the heating effect. A thermal effects, like bonds rupture by intense radiation [1,17], are also possible in the phase-change materials, but they are normally considered in the frame of reverse phase transition from crystalline to amorphous state. In the current study we are developing a proper thermal model to describe influence of a single femtosecond laser pulse on the thin film. The second process is material

melting of, since the fluence of laser pulses in our case is enough to provide its heating to the melting point and to supply the latent heat of melting. The last one is the material crystallization provoked by elevated temperature should be simulated to provide us with a physical picture leading to the appearance of the modified areas.

4.1. Temperature distribution induced by ultrashort laser pulse

Description of the process of heating semiconductors and dielectrics under the impact of femtosecond pulses is a complex theoretical problem. Here absorption of radiation occurs with interband transitions, which leads to the generation of free charge carriers. As their concentration increases, intraband absorption (absorption by free carriers) becomes significant. Electrons and holes absorb radiation, and eventually the carrier energy is transferred to the lattice through the electron-phonon interaction. However, electrons and holes energy distribution can be non-equilibrium and the concentration can change significantly during pulse impact. These factors are the main challenges for the analysis of the interaction of an ultrashort pulse with an absorbing medium.

In this work, we do not consider directly the process of excitation of the electron subsystem during the pulse impact, but focus on the final result in the form of local heating of a thin film. Nevertheless, we cannot completely ignore the mentioned above physical processes of non-equilibrium laser-matter interaction. We assume that the most significant contribution of excited electronic subsystem is increased with light absorption due to a large number of photo-generated charge carriers. Therefore, we estimated the value of the absorption coefficient, connected to free carriers using the Drude model for the harmonic oscillator [18]:

$$\alpha_f = \frac{N\lambda^2 q^2}{4\pi^2 \epsilon_0 m^* c^3 n \Delta t} \quad (1)$$

where N is the concentration of free carriers, λ is the free space laser wavelength, q is the fundamental charge, ϵ_0 is the vacuum permittivity, m^* is the conductivity effective mass of the free carrier, c is the light-speed, and n is the real component of the semiconductor refractive index, Δt is the mean time between collisions of the oscillating particles and the nuclei. The value of m^* for GST225 can be taken as 0.1 of the free

electron mass [19] and Δt can be estimated as 1 ps [20]. Charge carrier density can be estimated as [21]:

$$N \approx \frac{\alpha(1-R)I\tau}{\hbar\omega}, \quad (2)$$

where α is the linear absorption coefficient of the amorphous GST225, R is the reflection coefficient, τ is pulse duration, ω is laser radiation frequency and I is the peak pulse intensity.

The maximum beam intensity achieved with our equipment is of the order of 10^{11} W/cm². Tabulated absorption coefficient of non-excited amorphous GST225 for 1030 nm wavelength $\sim 10 \mu\text{m}^{-1}$ and reflectivity is 0.428 [22]. Thus, density of free charge carriers induced by femtosecond laser irradiation is about $5 \cdot 10^{21}$ cm⁻³. Hence, the rough order of magnitude estimation of free carriers' absorption is about $5 \mu\text{m}^{-1}$. In our calculations, we use the sum $\alpha_{\text{eff}} = \alpha + \alpha_f$ as the value of effective phenomenological absorption coefficient. To connect experimental results with such theory, we proceeded from the premise that the edge of the modified area presented itself the boundary of the molten region formed as a result of irradiation, which is reasonable to expect due to the previous results [9]. If so, the value of the effective absorption coefficient, which provides best match between experiment and simulation, is $16.2 \mu\text{m}^{-1}$ which is surprisingly close to the estimation given above.

Therefore we have to solve the problem of thermal conductivity for a GST225 amorphous film with a thickness of 230 nm and a lateral size 1 mm thermally contacting with substrates with different thermal parameters. We used a dielectric substrate with the thickness 1 μm and a conductive substrate with the thickness 0.5 μm . A conductive substrate, the actual-used substrate, was a multilayer structure made of metallic layers on top of the silicon base (see Section 2.1). However, within the range of simulation parameters examined in this study there is no need to perform calculations with such a complicated structure. Since we are interested in the heat dissipation into the substrate, but not in the distribution of heat in it, it is reasonable to replace multilayer structure with a thick homogeneous conductive layer. Trial calculations showed no difference in heat dissipation with such a replacement.

The film is heated by a single laser pulse with $\tau = 185$ fs and a $\lambda = 1030$ nm. The beam has spatial Gaussian profile; the beam radius is 30 μm . The pulse energy can take values from 200 to 1200 nJ.

Since an ultrashort pulse appears to be a heat source, we divide the heat conduction problem into two parts: film heating during $\tau = 185$ fs and the subsequent heat transition from the heated region into film bulk.

The solution is found by the finite difference method using an implicit scheme. Program for numerical modeling was written in Python version 3.9 using NumPy and SciPy libraries. Uniform mesh with coordinate increments 50 nm and 5 nm for r and z directions, respectively, was used. The constant time increment was 1 fs for the calculation of heating with a pulse during 185 fs and 0.5 ns for simulation of subsequent cooling.

4.1.1. Heating the film under the laser irradiation

The pulse intensity has an axisymmetric spatial distribution, which allows solving heat-transfer equation in the cylindrical coordinate system:

$$k \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T_0}{\partial r} \right) + \frac{\partial^2 T_0}{\partial z^2} \right) + q(r, z) = c\rho \frac{\partial T_0}{\partial t} \quad (3)$$

with initial condition:

$$T_0|_{t=0} = T_{\text{room}} \quad (4)$$

Here, $q(r, z)$ is the volumetric heating source, defined in accordance with the spatial and temporal distribution of light intensity in absorbing media:

используется α и α_{eff}

$$q(r, z) = (1-R) \frac{2\alpha E}{\pi w^2 \tau} \exp\left(-\frac{r^2}{w^2}\right) \exp(-\alpha_{\text{eff}} z), \quad (5)$$

where R is the reflection coefficient, E is the pulse energy and w is the radius of the Gaussian beam. The last exponential factor reflects the Beer law, which includes the absorption coefficient α_{eff} , calculated in the beginning of Section 4.1.

Film is brought into thermal contact with a substrate with different thermal properties. Continuity conditions at the interface between film and substrate requires:

$$k \frac{\partial T}{\partial z}|_{z=h} = k_{\text{substrate}} \frac{\partial T_{\text{substrate}}}{\partial z}|_{z=0}, \quad T|_{z=h} = T_{\text{substrate}}|_{z=0}, \quad (6)$$

where h is a film thickness.

$T_{\text{substrate}}|_{z=0}$ and $\frac{\partial T_{\text{substrate}}}{\partial z}|_{z=0}$ can be found by solving the heat equation for substrate:

$$a_s \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T_{\text{substrate}}}{\partial r} \right) + \frac{\partial^2 T_{\text{substrate}}}{\partial z^2} \right) = \frac{\partial T_{\text{substrate}}}{\partial t}, \quad (7)$$

$$T_{\text{substrate}}|_{z=h_s} = T_{\text{room}}, \quad \frac{\partial T_{\text{substrate}}}{\partial r}|_{r=R_N} = 0, \quad \frac{\partial T_{\text{substrate}}}{\partial r}|_{r=0} = 0,$$

where h_s is a substrate thickness and R_N is a lateral size of the film.

Radial symmetry condition:

$$\frac{\partial T_0}{\partial r}|_{r=0} = 0, \quad (8)$$

The lateral size of the film is significantly larger the laser beam radius; therefore, the condition on the film lateral boundary is:

$$\frac{\partial T_0}{\partial r}|_{r=R_N} = 0, \quad (9)$$

At the first stage of the solution, we render the top film surface impervious to heat, because any possible heat removal in this case is negligible in comparison with the energy furnished to the film by the laser pulse. Therefore, for the film surface:

$$\frac{\partial T_0}{\partial z}|_{z=0} = 0, \quad (10)$$

The heating process causes melting of the amorphous phase; i.e. is accompanied by absorption of the melting latent heat. In this case, an equivalent specific heat method can be used taking into account such absorption [1]. The latent heat of melting ΔH_f is converted into appropriate units for specific heat c_E and added to the specific heat term over a temperature range at which melting occurs. The equivalent specific heat which includes the latent heat term and varies over the solidification range has the following form:

$$c_E = c + \Delta H_f \frac{df_m}{dT}, \quad (11)$$

where f_m is the temperature dependent fraction of the melt which has the form of a sigmoid function.

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4.1.2. Heat spread from a heated area

We solve the problem of thermal conductivity for a locally heated film brought into thermal contact with a substrate with different thermal properties. For the film, the equation has the form:

$$k \left(\frac{1}{r} \frac{\partial}{\partial r} \left(r \frac{\partial T}{\partial r} \right) + \frac{\partial^2 T}{\partial z^2} \right) = c\rho \frac{\partial T}{\partial t} \quad (12)$$

with initial condition:

$$T|_{t=0} = T_0. \quad (13)$$

For the film-air interface:

$$k \frac{\partial T}{\partial z} \Big|_{z=0} = a(T - T_{room}). \quad (14)$$

Conditions for $z = h$ and $r = 0$; R_N are the same as in (6) and (8), (9), respectively.

Temperature transfer in the bulk of the film can also cause melting of the amorphous phase. Then this also has to be taken into account by introducing the equivalent specific heat, as shown in formula (11).

Numerical values of thermodynamic parameters for GST225 amorphous film are listed in Table 1.

4.2. Crystallization model

Due to statistical fluctuations, small regions of the crystalline phase constantly appear and disappear within the amorphous material. They can append monomers (atoms or molecules), which make them grow, or lose them, dissolving. The larger those regions are, the longer they exist. There is a critical size — once a cluster reaches it, the probability of dissolution becomes less than the probability of growth. Those critical clusters are the nuclei of crystalline phase. As the growth of clusters proceeds through a sequence of steps of joining monomers due to fluctuations, the process of nucleation is by no means instantaneous. Kinetics of this process is described in terms of time-lag Θ for nucleation, which is a characteristic time of reaching the steady state value of nucleation rate J_{ss} (rate of nuclei production per unit volume of initial phase). Growth of nuclei in a condensed phase is controlled by transition of monomers across the interface of a crystalline cluster mainly. Under this assumption, the growth rate U (the rate at which the interface of a crystalline cluster advances) can be considered as a constant [36]. These three quantities Θ , J_{ss} , and U fully describe the crystallization at a given

semiconductors [31,39].

The rate D_n of monomers attachment to the spherical cluster of n monomers is defined by Kelton et al. [40] as

$$D_n = 4n^{\frac{2}{3}}\gamma \exp\left(-\frac{\Delta G_{n+1} - \Delta G_n}{2k_B T}\right), \quad (18)$$

where G_n is the free energy of a cluster of n monomers:

$$\Delta G_n = (6\sqrt{\pi} v_m n)^{\frac{2}{3}} \sigma - n\mu_m. \quad (19)$$

The value of γ defines the unbiased molecular jump frequency (Hz) at the cluster surface. It can be expressed in terms of diffusion coefficient D [40]:

$$\gamma = \frac{6D}{\lambda_j^2}, \quad (20)$$

where λ_j — average jump distance, which could be estimated as $\sqrt[3]{v_m}$. Diffusion coefficient is related to the viscosity η via Stokes-Einstein equation

$$D = \frac{k_B T}{3\pi\lambda\eta}. \quad (21)$$

The value of η is itself temperature dependent. To model this dependence, we adopt the formulation proposed by Aladool et al. [29]. They assumed, that while below the glass transition temperature T_g the temperature dependence of η obeys simple Arrhenius behavior, at higher temperatures it is more complex and is described by MYEGA (Mauricio-Yue-Ellison-Gupta-Allan) model [41], leading to:

$$\log \eta = \begin{cases} 12 + \log \left(\exp \left[\frac{E_\eta}{k_B} \left(\frac{1}{T} - \frac{1}{T_g} \right) \right] \right) & T < T_g, \\ \log \eta_\infty + (12 - \log \eta_\infty) \frac{T_g}{T} \exp \left[\left(\frac{m}{12 - \log \eta_\infty} - 1 \right) \left(\frac{T_g}{T} - 1 \right) \right] & T \geq T_g. \end{cases} \quad (22)$$

temperature and should be considered while constructing a TTT diagram.

Kashchiev [37] obtained the following expression for the nucleation time-lag:

$$\Theta = \frac{4n_c k_B T}{\mu_m D_c}, \quad (15)$$

where k_B is the Boltzman constant, μ_m is the difference (positive) in Gibbs free energy per one monomer transition from initial to transformed phase, and D_c is the rate (Hz) of monomer addition to the spherical cluster of critical size n_c :

$$n_c = \frac{32\pi}{3} \frac{\sigma^3 v_m^2}{\mu_m^3}, \quad (16)$$

where σ is the interfacial energy, v_m is the volume of material per one monomer. We consider the monomer to be an arbitrary atom of material composition. Therefore, the value of v_m can be estimated from the density of GST225 and its average molar mass.

Free energy change during crystallization can be approximated using Thompson-Spaepen formula [38]:

$$\mu_m = v_m \Delta H_f \frac{T_m - T}{T_m} \frac{2T}{T + T_m}, \quad (17)$$

where ΔH_f is the enthalpy of fusion, and T_m is the melting point. This approximation is considered appropriate for the case of chalcogenide

The glass transition temperature is defined here as the temperature at which the viscosity equals 10^{12} Pa s. Parameters of MYEGA model are extrapolated infinite temperature viscosity η_∞ and fragility index m . In our simulation, index m is selected by matching TTT curves with the corresponding cooling curves, as described in the following section. At temperatures below T_g , the viscosity follows Arrhenius behavior with activation energy E_η .

The steady state nucleation rate is described by the equation [37,40]:

$$J_{ss} = v_m^{-1} 4n_c^{\frac{2}{3}} \gamma \sqrt{\frac{\mu_m}{6\pi k_B T n_c}} \exp\left(-\frac{\Delta G_c}{k_B T}\right), \quad (23)$$

where ΔG_c is the Gibbs free energy of the critical cluster:

$$\Delta G_c = \Delta G_{n_c} = \frac{16\pi\sigma^3 v_m^2}{3\mu_m^2} = \frac{\mu_m n_c}{2} \quad (24)$$

Growth rate can be defined as [42]:

$$U = \frac{4k_B T}{3\pi\lambda^2\eta} \left(1 - \exp\left(-\frac{\mu_m}{k_B T}\right) \right). \quad (25)$$

Once $\Theta(T)$, $J_{ss}(T)$, and $U(T)$ are established, the TTT curve could be calculated. Starting from the usual Kolmogorov-Johnson-Mehl-Avrami time dependence of crystallinity X_c under condition of bulk nucleation [16]:

$$\ln(1 - X_C) = -\frac{4}{3}\pi U^3 \int_0^t J(t')(t-t')^3 dt' \quad (26)$$

and using the following expression for time-dependent nucleation rate [37]:

$$J(t) = J_{ss} \left[1 + 2 \sum_{n=1}^{\infty} (-1)^n \exp\left(-n^2 \frac{t}{\tau_l}\right) \right], \quad (27)$$

we eventually arrive at:

$$\begin{aligned} \ln(1 - X_C) = & -\frac{\pi}{3} J_{ss} U^3 \tau_l^4 \left[\left(\frac{t}{\tau_l}\right)^4 - \frac{2\pi^2}{3} \left(\frac{t}{\tau_l}\right)^3 + \frac{7\pi^2}{30} \left(\frac{t}{\tau_l}\right)^2 - \frac{31\pi^2}{630} \left(\frac{t}{\tau_l}\right) + \right. \\ & \left. + 48 \sum_{n=1}^{\infty} \left(\frac{(-1)^n}{n^8} \left(\exp\left[-n^2 \frac{t}{\tau_l}\right] - 1\right)\right) \right], \end{aligned} \quad (28)$$

where $\tau_l = 6/\pi^2 \Theta$.

Provided that the value of X_C is given, the crystallization time t could be calculated solving the above equation numerically. The kinetic parameters of amorphous GST225 used in the simulation could be found in Table 1. The procedure for constructing TTT curves is described in subsection 5.1.

In order to determine the optical contrast on the surface of the formed crystalline structure, the corresponding distribution of the reflection coefficient over the surface was calculated using the transfer matrix method (TMM) code [43]. For this purpose, the complex dielectric function of partially crystalline material ($X_C < 1$) was determined according to effective medium theory as [4]:

$$\begin{aligned} \varepsilon(X_C, \varepsilon_a, \varepsilon_c) = & \frac{1}{4} \left[2\varepsilon_p - \varepsilon'_p + \sqrt{(2\varepsilon_p - \varepsilon'_p)^2 + 8\varepsilon_a \varepsilon_c} \right], \\ \varepsilon_p = & (1 - X_C)\varepsilon_a + X_C \varepsilon_c, \\ \varepsilon'_p = & (1 - X_C)\varepsilon_c + X_C \varepsilon_a, \end{aligned} \quad (29)$$

where ε_a and ε_c are complex dielectric constants of amorphous and crystal GST225.

5. Discussion

5.1. TTT diagram and critical cooling rate

In our assumption, for the composition of GST225 under constant other conditions, crystallinity X_C is determined by the material cooling rate. In Fig. 3, the lines of gray shades show TTT curves corresponding to different values of crystallinity (labeled on the curves). Equations (7) and (12) allow the construction of cooling curves at any point of the thin film – curves constructed for spots edges are shown in orange and blue in Fig. 3. The crystallinity formed during particular cooling regime is determined by the TTT curve with which the cooling curve has a single intersection point. The value of cooling rate required for the formation of a phase with a given crystallinity is defined as the slope of a line tangent to this intersection point on the corresponding TTT curve.

All the parameters necessary to plot a diagram are known from the literature (their values and corresponding sources are given in Table 1). However, for fragility index m , its value varies over a wide range in different sources, and the kinetic model is very sensitive to it. So, it was required to adjust this parameter during construction of TTT diagram.

The procedure of TTT curve plotting consisted of two stages. At the first stage, according to the results of solving the problem of thermal conduction, the cooling curve was determined at the point corresponding to the edge of the bright spots. The latter was determined from

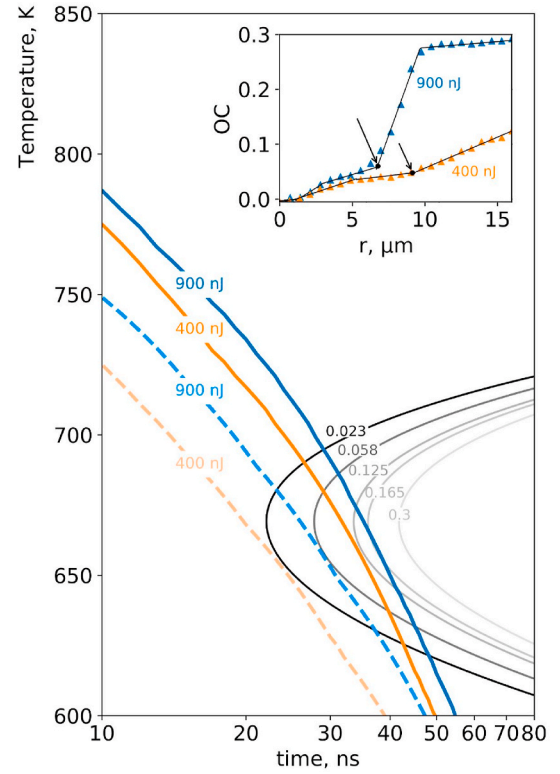


Fig. 3. TTT diagram at various crystallinity (as indicated) and cooling traces for edge of crystalline spots (solid lines) and pale spots (dashed lines), obtained for different pulse energies; OC profiles are shown on the inset: experimental data (dots) and its approximation with broken line (solid line); arrows indicate points at which $T(t)$ curves for crystalline spots were measured.

the optical contrast profile, as shown in the inset in Fig. 3. The minimum optical contrast of the bright spot was determined at same point as well. Finally, using TMM code and eq. (29), we calculated crystallinity X_C corresponding to the edge of the bright spot. For this stage we used optical images of spots, obtained for the film on substrate I. Further verification will show that calculated diagram is correct for both different substrates and different film thickness and is therefore sensitive to the GST225 properties solely.

The second stage was to calculate the TTT curve for a given crystallinity value with the use of (28). The TTT curve has to be tangent to the corresponding cooling curve. To achieve this, we selected an appropriate value of fragility index m .

Procedure was repeated for the spots obtained at different pulse energies to check the accuracy of fragility index in order to make sure that the selected parameter is correct. For the simulation $m = 67$ The best match between two types of curves was shown for the simulation with $m = 67$ Using the given fragility index, “C”- shaped curves, that define crystallization temperature, crystallization time and cooling rate required for different levels of crystallinity, were plotted, (see Fig. 3). The “C” curve divides the TTT diagram into three parts, starting from glass transition temperature up to the melting point. Upon cooling from the melting point increased supercooling provides the driving force for crystallization process, which is modeled by eq. (17) Consequently, cooling leads to higher rate of crystallization process and therefore to a decrease of crystallization time. At low temperatures, however, crystallization time increases meaning decrease of crystallization rate. It is explained by decreasing atomic mobility, which is described by eq. (22). Ultimately, crystallization process is governed by the interplay between two abovementioned factors having opposite temperature dependencies, which lead to “C”-shaped form of TTT diagrams. At the bottom, the “C”-shaped curve is limited by $T = 423$ K – the equilibrium

crystallization temperature. Corresponding cooling rate in this case is about 10 K/min, which is consistent with reported experimental data [13,44]. The second region is situated from 775 K to T_m , where nucleation is inhibited by too high atomic diffusion. Between these areas, there is the so-called nose of the TTT curve which determines the minimum crystallization time and the corresponding temperature.

According to the simulation results, the nose of TTT diagram is located at 670 K and does not depend on crystallinity. If one assumes the lower detectable limit of crystallization $X_c = 10^{-8}$, the minimum time required to reach a given crystallinity is $t_c = 1$ ns. Critical cooling rate is defined as the slope of a tangent from T_m to the nose point, and it gives a value of 316 K/ns. Values of TTT nose position, incubation time and critical cooling rate are within the range of values reported in Refs. [45, 46].

Fig. 3 represents pairs “cooling curve-TTT curve” for the pale and bright spots formed after 400 and 900 nJ laser pulses. Pale spots have the form of barely crystallized material with a crystallinity about 0.02–0.06. The cooling rate required for their formation is about 18 K/ns, which is also in line with the previously reported experimental data [47]. For the bright spots formation with a crystallinity at least about 0.1 the cooling rate should not exceed 6.4 K/ns.

To achieve the full crystallization, the cooling rate must be ensured to be no more than 3.8 K/ns. This requires reducing the heat dissipation from the film. The possible options to achieve this effect are the following: to use the substrate with a lower thermal conductivity or/and to increase the film thickness.

5.2. Temperature distribution and crystallization

As mentioned above, impact of 185 fs pulses is considered as instantaneous heating of the material above melting point. After the end of the pulse, the irradiated region begins to cool down and solidify. According to the simulation, pulse influence leads to heating of the irradiated area to temperatures above the melting point and local melting in the central part. The molten area turns out to be heated above 1000 K and occupies several nanometers of the film bulk (Fig. 4, b). Cooling curves at various points on the surface and in the depth of the irradiated region are shown in Fig. 4, c. The moment of time when the crystallization process begins is taken as $t = 0$. For heated melted

regions, $t = 0$ corresponds to the moment when the material has cooled down to the melting temperature of 890 K. For the regions, heated below the melting point, the moment of the end of the pulse is taken as the origin of a time scale.

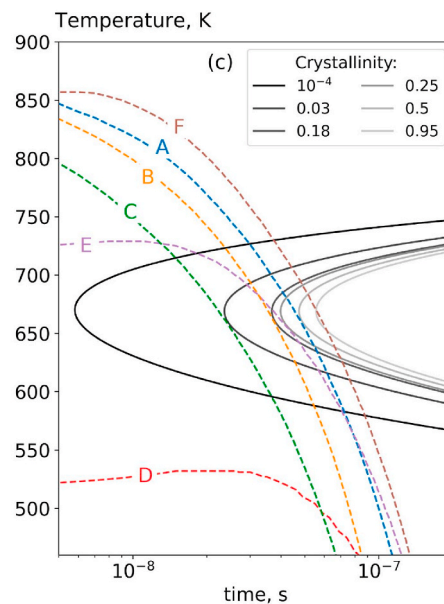
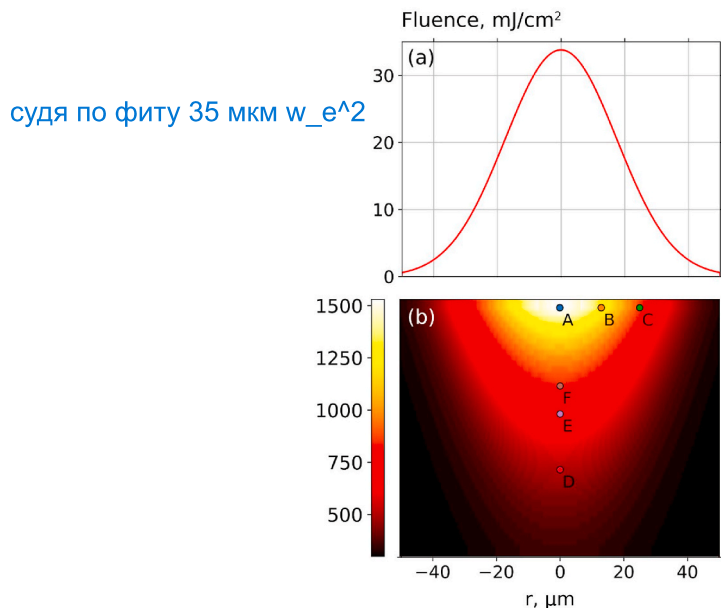
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The initial temperature to which the material was heated together with the subsequent cooling rate strongly affects the obtained crystallinity. The method described above for converting the cooling rate into crystallinity makes it possible to determine the value of the latter at different points of the irradiated material, relying on the cooling curves obtained by numerically solving the heat equation. Fast cooling from the melting temperature leads to the transformation into the glass containing a small amount of the crystalline fraction – about 0.03 (curve C, Fig. 4 (c)), and the pale spot formation is observed. The slower cooling process, the more crystalline nuclei can be formed. If the cooling rate of the melt decreases to 6.4 K/ns, one can observe the formation of a bright center with crystallinity of about 0.1 and higher (curves A, B, Fig. 4 (c)).

However, the crystallization occurs not only from the melt. In deeply located layers of a material, temperature of about 670 K can be maintained for longer than 60 ns. This, in turn, leads to a transformation from solid amorphous to solid crystalline material (curves E, F, Fig. 4 (c)). As a result, the deep-lying thin layers with a high value of X_c are formed.

In even deeper layers, crystallization is impossible due to fast cooling from low (about 500 K, see curve D in Fig. 4 (c)) temperatures. Higher temperatures cannot be achieved due to insufficient absorption and energy confinement at the surface due to melting.

Based on the TTT diagram, we found the distribution of crystalline fraction in the bulk of the material on substrate I formed after irradiation with pulses of different energies (Fig. 4). The pulse energy being increased, the crystallinity in the center of the irradiated region also increases. Thus, the crystallinity in the near-surface layers for 350, 450 and 650 nJ pulses is 0.34, 0.74 and 0.83, correspondingly. Moreover, with an increase in the pulse energy, more and more deeply located layers are heated to the temperature required for crystallization. Consequently, the thickness of the region occupied by the crystalline fraction increases. As the final result, the optical contrast of the surface increases as well. To make sure the modeling results are correct we calculated the surface optical contrast distribution from simulated structures, and as shown in Fig. 5 a reasonable agreement between



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Fig. 4. Image (b) shows the temperature distribution in the bulk of thin film induced by the single pulse with fluence profile shown on the graph (a) (pulse energy 650 nJ); diagram (c) demonstrates the cooling curves at the points marked on (b) and corresponding to the cooling rates crystallinity, according to the TTT diagram in Fig. 3.

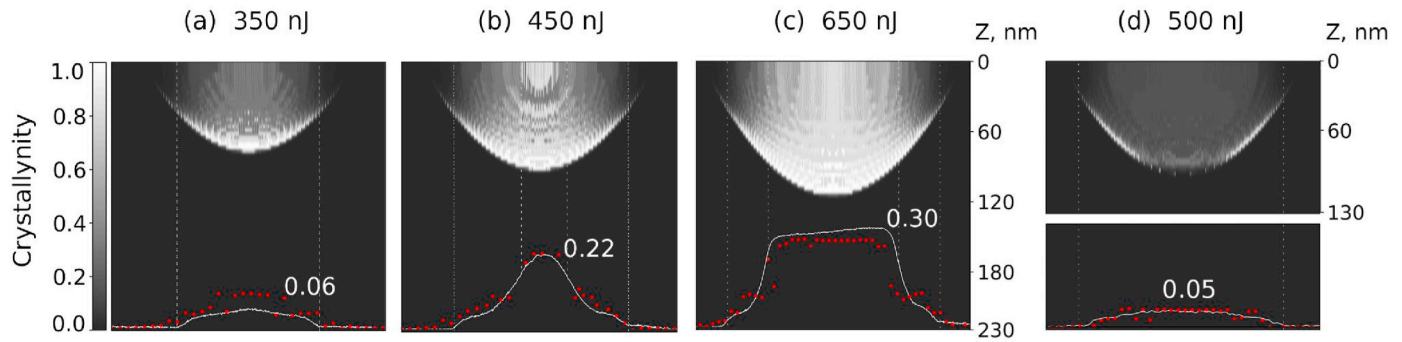


Fig. 5. Calculated distribution of crystalline fraction in the film bulk after irradiation with a single pulse for the films of 230 nm thickness (a–c) and 130 nm thickness (d). The calculations were performed for three different pulse energies, which are indicated above the corresponding image. Inserts show a comparison between the optical contrast from simulated structures (red dots) and experimental data (white solid line) from Fig. 1; the maximum contrast values are indicated. Lateral size of the images is 60 μm .

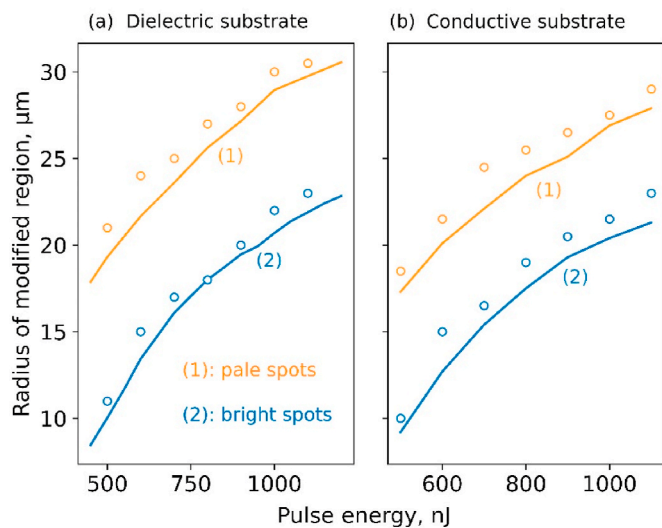


Fig. 6. Dependences of the radius of the modified region on the incident pulse energy for substrate I (a) and II (b). The solid curves are experimental data, and dots are simulation results.

numerical and experimental results is observed.

An increase in the pulse energy entails an increase of the modified areas radii. This corresponds to the fact that the phase transformation is a threshold process. In GST225 films crystallization with Gaussian laser pulses occurs only upon reaching a certain level of the energy fluence — the threshold fluence of crystallization [9]. The experimentally observed increase of the radii of the modified areas can be traced in the same way from the simulation results. As seen in Fig. 6, the calculated values reproduce the experimental data for both substrate types quite well — deviation is not more than 5%. The same procedure was carried out for substrate II, and the same agreement between theory and experiment was achieved.

Fig. 6 demonstrates that the sizes of the modified regions obtained using conductive and dielectric substrates are very close to each other. At the first glance this fact may seem to be counterintuitive since conductive substrate is supposed to provide effective heat removal from the film and thereafter higher cooling rate. As a result, the sizes and crystallinity of the modified regions are supposed to be significantly smaller for the film on the conductive substrate. Yet the experiment reveals little difference in these parameters for the dielectric substrate. The developed herein model for the laser-induced phase transitions, being successful in simulating the experimental data, suggest satisfactory explanation for this dilemma.

The model is developed upon the interplay between two kinetic processes: heat conduction and crystallization. Substrate conductivity type plays a crucial part only when the crystallization time is longer than the time taken for heat to reach the substrate. Using definition of the thermal diffusion length $\delta = 2\sqrt{\alpha t}$, we could estimate the characteristic time of heat diffusion ca. 70 ns through the film of 230 nm thickness. The characteristic timescale of crystallization according to our simulations demonstrates approximately the same value, as shown in Fig. 3. Therefore, when the heat reaches the substrate the phase transition is already over. Under such conditions the heat conductivity of the substrate is of minimal importance.

Ultimately, the relationship between dynamics of crystallization and heat conductivity defines the role of substrate in overall process. As we have chosen a relatively thick film, it happens that in our experiment this role is insignificant. For thinner films thermal diffusion time becomes shorter than the characteristic crystallization time, and the substrate type will be determining.

The simulation was performed for a 130 nm thick film as well (Fig. 5 (d)). Initial assumption that proper heat flow conditions can be achieved only at a sufficiently thick film is confirmed. Indeed, at thick film of 130 nm the minimal cooling rate was about 9 K/ns, which led to the solidification of the melt with resulting crystallinity value of about 0.1. Therefore, the optical images of this structure are presented as the pale spots with low optical contrast, which is observed in the experiment (see Fig. 1).

6. Conclusions

In this paper we demonstrated the crystallization triggered by a single femtosecond laser pulse in $\text{Ge}_2\text{Sb}_2\text{Te}_5$ thin film. Solving the set of equations describing the temperature dependencies of the incubation time, nucleation rate and crystal growth we determined the temperature regime required for crystallization in terms of TTT diagrams. We solved the heat conductivity problem to find the heat distribution induced by the single femtosecond laser pulse and apply the calculated TTT diagrams to determine the amount of formed crystalline fraction. Comparing the experimental data with the simulation results, we have found the range of cooling rates at which crystallization occurs. It was determined that the formation of the lower detectable limit of crystallinity 10^{-8} requires cooling rate of 316 K/ns and full crystallization can be achieved at cooling rate 3.8 K/ns.

Based on simulated results, we defined the distribution of crystalline fraction in the bulk of the material formed after irradiation with pulses of different energies. Optical contrast from simulated structures agreed with experimental results well.

Appropriately low cooling rate, at which crystallization process is possible in our experiments, was achieved for sufficiently thick films only. Little difference is observed in results of laser induced modification

whether the substrate is a good or a poor heat conductor. On the basis of simulation results, it is explained by finite time of heat diffusion through the film, which exceeds the crystallization time for the thick films. . However, simulation shows that the complete crystallization of the irradiated region may not be achieved for a 230 nm thick film on substrate with low thermal conductivity. For that purpose, it is needed to reduce heat transfer from the irradiated region by increasing the film thickness or, presumably, covering on top of the GST film.

CRedit authorship contribution statement

Tatyana Kunkel: Conceptualization, Methodology, Investigation, Data curation, Writing – original draft, Writing – review & editing. **Yuri Vorobyov:** Conceptualization, Methodology, Investigation, Writing – original draft. **Mikhail Smayev:** Methodology, Investigation, Writing – original draft, Writing – review & editing. **Petr Lazarenko:** Resources, Methodology, Investigation, Writing – original draft. **Alexey Romashkin:** Methodology, Validation. **Sergey Kozyukhin:** Conceptualization, Methodology, Supervision, Project administration, Writing – original draft, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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